Ankita Das

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.38
BILTHAI H.S SCHOOL	Class XII (TRIPURA BOARD OF SECONDARY EDUCATION)	2020	65.00%
BILTHAI H.S SCHOOL	Class X (TRIPURA BOARD OF SECONDARY EDUCATION)	2018	69.80%

SKILLS

Programming Languages: Python, C, MATLAB, SQL

AI / ML: Deep Learning (CNN, U-Net), Machine Learning, Statistical Modelling, Image Segmentation, Object Detection, Reinforcement Learning

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

Software & Tools: VS Code, PyCharm, Excel, Jupyter Notebook, ITK-SNAP, MIMICS, GROMACS

COURSEWORK

Artificial Intelligence in Healthcare, Statistical Inference Methods in Bioengineering, Medical Image Processing, Ultrasound in Medicine, Biomedical Devices

PROJECTS

Spatially Adaptive U-Net for 3D Breast Tumor Segmentation

MTech Thesis

- Trained a **U-Net deep learning model** in PyTorch for breast tumor segmentation using public ultrasound datasets.
- Analyzed model performance on **our clinical 3D B-mode ultrasound data** across different tumor positions using multiple evaluation metrics.
- Visualized and compared model results using heatmaps; currently exploring **BI-RADS classification using ResNet-50**.

Automated Brain Aneurysm Detection & Classification

BTech Thesis

- Developed an **AI**-based algorithm for automated brain aneurysm detection and classification using **deep learning** techniques on CT angiography images.
- Performed manual segmentation of aneurysm regions under expert supervision to create ground-truth labels for model training.
- Designed and trained a **Convolutional Neural Network (CNN) model**, achieving **96%** accuracy in detecting aneurysms.
- Used a Random Forest algorithm to classify aneurysm types (ruptured/unruptured, saccular, fusiform, fusisaccular).

INTERNSHIPS

Data Visualization Associate Intern -- Excelerate

- Working with **PostgreSQL and structured datasets** for data analysis and reporting.
- Performing **SQL**-based data exploration, cleaning, integration, and validation to prepare **datasets** for analysis.
- Creating **visualizations** and **dashboards** in Looker Studio to analyze trends, performance, and key metrics.
- Generating **data-driven insights** and recommendations and presenting findings through reports and dashboards.

Project Intern - CSIR-National Chemical Laboratory, Pune

Physical & Materials Chemistry Division

- Evaluated conformational ensembles of monomeric -synuclein- a protein whose aggregation is linked to Parkinson's Disease -using molecular dynamics simulations (GROMACS).
- Characterized how the protein's conformational ensembles evolve as a function of varied salt (ionic strength) concentrations, applying statistical modelling to interpret the dynamics.
- Findings contributed to a peer-reviewed study on mutation- and ionic-strength-driven -synuclein aggregation, later published in ACS Chemical Neuroscience (2026).

PUBLICATIONS

Tammara V., **Das A**., Desai Z., Das A. "Combined Effects of Mutation and Ionic Strength on -Synuclein Reveal Generic Features of Aggregation-Prone Monomeric States." ACS Chemical Neuroscience, 2026, 17(14), 2752-2770. Impact Factor: 4.5

DOI: https://doi.org/10.1021/acschemneuro.6c00439

AWARDS & ACHIEVEMENTS

- **1st Prize - UG Category Poster Presentation,** 7th ANSACON 2025, NEIGRIHMS Shillong (April 2025) - Automated Brain Aneurysm Detection & Classification.

- Design Patent No. 470694-001 granted by The Patent Office, Government of India (Aug 2025) for Automatic Medical Device for Detection and Classification of Brain Aneurysm.